University of Florida, Department of Mathematics

Probability PhD Exam, May 1989

1. Let X_n be independent identically distributed random variables and $S_n = \sum_{i=1}^n X_i$. Suppose

$$\mathrm{E}\left[\sup_n \left|\frac{S_n}{n}\right|\right] < \infty.$$

Show that

$$\mathrm{E}\left[|X_1| \log^+ |X_1|\right] < \infty,$$

where

$$\log^+ x = \begin{cases} \log x, & x \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Hint: $C_n = \prod_{j=1}^n \mathrm{P}[|X_j| \leq j] \to c > 0$. If $T = \inf\{n : |X_n| > n\}$, then

$$\sum_{n=1}^{\infty} \frac{1}{n} \int_{\{T=n\}} |X_n| \, dp = \sum_{n=1}^{\infty} \frac{C_{n-1}}{n} \int_{\{|X_1|>n\}} |X_1| \, dp$$

and

$$\sum_{n=1}^{\infty} \frac{1}{n} \int_{\{T=n\}} |S_{n-1}| \, dp < \infty.$$

2. Let X_n be independent Poisson variables. Suppose $\sum_{n=1}^{\infty} X_n < \infty$ a.s. Show that $\sum_{n=1}^{\infty} \mathrm{E}[X_n] < \infty$. Hint. Borel–Cantelli.

3. This problem has two parts.

a) Define the notion of conditional expectation. In particular explain the notion $\mathrm{E}[X \mid Y]$ where X and Y are random variables.

b) Suppose X, Y are random variables such that $\mathrm{E}(X^2), \mathrm{E}(Y^2) < \infty$ and

$$\mathrm{E}[X \mid Y] = Y, \qquad \mathrm{E}[Y \mid X] = X.$$

Show that $X = Y$ a.s.

4. This problem has three parts.

a) A sequence of characteristic functions on $\mathbb{R}^1$ converging pointwise to a characteristic function does so uniformly on compact sets. Explain why - no need to give complete proofs.

b) Let $b_n \in \mathbb{R}$ and f a characteristic function not identically equal to 1. Suppose $f(b_n t)$ converges to a characteristic function. Show that $\lim b_n = b \in \mathbb{R}$.

c) Let $\phi(t) = \frac{1}{8}[1 + 7e^{it}]$. Show that ϕ is a characteristic function but that $|\phi|$ is not.
Hint. $|\phi|^2$ is the characteristic function of a measure concentrated on $\{-1, 0, 1\}$. If $|\phi|$ is the characteristic function of a measure m, then $|\phi|^2$ is the characteristic function of $m * m$.

5. Let X_t be a standard Brownian motion.

(i) Show that (X_t), $(X_t^2 - t)$, and $e^{uX_t - \frac{1}{2}u^2 t}$ are all martingales.

(ii) Let $T_a(w) = \inf\{t : X_t(w) = a\}$ for $a > 0$, the inf being defined to be $+\infty$ if the set is empty. Using (i), compute $\mathrm{E}[e^{-\lambda T_a}]$ for $\lambda > 0$, and show that $T_a < \infty$ a.s. and $\mathrm{E}[T_a] = \infty$.

6. State and sketch the proof of the martingale convergence theorem.

7. Using the martingale convergence theorem prove the following:

Let F_n be an increasing sequence of σ-fields, X_n a sequence of random variables dominated by an integrable random variable Y: $|X_n| \leq Y$. Suppose $X_n \to X$ a.s. Then

$$\mathrm{E}[X_n \mid F_n] \to \mathrm{E}[X \mid F_\infty] \quad \text{a.s.}$$

Hint. Put the

$$U_n = \sup_{m \geq n} |X_m - X|.$$

Then for $m \geq n$,

$$\mathrm{E}[|X_m - X| \mid F_m] \leq \mathrm{E}[U_n \mid F_m].$$

$$\mathrm{E}\left\{\limsup_m \mathrm{E}[|X_m - X| \mid F_m]\right\} \leq \mathrm{E}\{\mathrm{E}[U_n \mid F_\infty]\} = \mathrm{E}[U_n].$$